ClinicalTrials.gov

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Constructing Complex Search Queries

Introduction

This page explains the use of search operators, terms, syntax, and expressions to do an advanced search of study record data on ClinicalTrials.gov.

Search expressions consist of a sequence of terms and operators. The operators affect which studies are returned and how they are ordered in retrieval sets by changing how the search terms are contextualized or interpreted. Search terms are words or phrases that must appear as values in the study records returned by the search. Users can create a search expression on the Expert Search page found in the main menu under Find Studies and on the Site Map. In addition, users can create and edit a search expression by executing a search using Basic Search and clicking the Expert Search link button in Focus Your Search.

Search syntax refers to the arrangement of search terms and operators in search expressions used for evaluation by the search engine.

Search operators

In general, operator precedence (from highest to lowest) is as follows: search terms and source operators, the NOT operator and context operators, the AND operator, and the OR operator.

The following operator types are described in the sections below:

- Boolean operators (https://clinicaltrials.gov.#booleanOpr)
- Grouping operators (https://clinicaltrials.gov.#groupOpr)
- Context operators (https://clinicaltrials.gov.#contextOpr)
- Source operators (https://clinicaltrials.gov.#sourceOpr)
- Scoring operators (https://clinicaltrials.gov.#scoringOpr)

Boolean operators

Boolean operators connect search terms and define their logical relationship.

Feedback

Operator	Example	Description
OR	youth OR teen	A binary operator used to retrieve study records containing either the left or right subexpression, or both. The weighted score for determining rank order in retrieval sets will be higher when both the left and right subexpressions are present in a study record.
AND	heart AND attack	A binary operator used to retrieve study records containing both the left and right subexpressions.
NOT	bethesda NOT maryland	A unary operator that acts only on the right subexpression. It is used to retrieve study records that do not contain the right subexpression.

Grouping operators

Grouping operators group search terms or operators in a query.

Operator	Example	Description
" "	"back pain"	Forces a sequence of words to be treated as a phrase.
()	(acetaminophen OR aspirin) AND NOT (heart failure OR heart attack)	Used to increase operator precedence in a search expression.

Context operators

Context operators, which control how search terms are evaluated, are immediately followed by parameters in square brackets without any spaces. All context operators have the same precedence as the NOT operator and apply to the subexpression on their right, which immediately follows the right square bracket without a space in between.

Operator	Example	Description
COVERAGE	COVERAGE[FullMatch]pain	Declares the degree to which a search term needs to match the text in an API field. There are four choices: • FullMatch—The search term must match all of the text in the field searched. • StartsWith—The search term must match the beginning of the text in the field searched. • EndsWith—The search term must match the end of the text in the field searched. • Contains—The search term must match part of the text in the field searched.
EXPANSION *	EXPANSION[None]SLE	Declares the degree to which a search term may be expanded. There are five choices: • None—The term is searched for exactly as is. Case and accent marks are significant, but consecutive spaces are treated as a single space. • Term—Similar to None but includes simple lexical variants such as plurals, possessives, alternate spellings, and compound words; ignores case, hyphens, and accent marks. • Concept—Similar to Term but includes synonyms based on the Unified Medical Language System (UMLS). Also has a slight scoring

Operator	Example	Description
		penalty, ranking any records that include the search terms higher than records that include only synonyms. • Relaxation—Similar to Concept but relaxes adjacency requirements so that partial terms are matches (e.g., a search for *heart disease* will return records with *heart* in one place and *disease* in another, as in the phrase "heart and lung disease"). Also has a significant scoring penalty, ranking any records that include the full search terms higher than records that include only partial terms. • Lossy—Similar to Relaxation but allows for missing partial terms (e.g., a search for *heart disease* will return records with *heart* but not *disease* and records with *disease* but not *heart*).
AREA	AREA[InterventionName]aspirin	Declares which search area should be searched. Search areas are defined on the ClinicalTrials.gov Search Areas (https://clinicaltrials.gov/data-api/about-api/search-areas) page. In addition to specifying search areas, it is possible to specify a field from the study structure (https://clinicaltrials.gov/data-api/about-api/study-data-structure). Any field from the study structure is searchable.

Operator	Example	Description
SEARCH **	heart attack AND SEARCH[Location](AREA[LocationCity]Portland AND AREA[LocationState]Maine)	Declares which subsection of the study structure (https://clinicaltrials.gov/data-api/about-api/study-data-structure) should be searched. For example, this operator can be used to search for a city and state within the same location (e.g. Portland, Maine), and exclude cities and states outside of that location (e.g. Portland, Oregon, and Augusta, Maine).

* The EXPANSION operator interacts with other operators as follows:

- Enclosing a sequence of words in quotation marks forces those words to be treated as a phrase and gives the phrase higher precedence than EXPANSION. As such, a quoted phrase following EXPANSION[Relaxation] or EXPANSION[Lossy] is effectively the same as EXPANSION[Concept].
- COVERAGE[FullMatch], COVERAGE[StartsWith], and COVERAGE[EndsWith] can be used only with EXPANSION[Concept], EXPANSION[Term], or EXPANSION[None].

** The SEARCH operator restricts search expressions to fields within a data element so that multiple pieces from that data element can be found together. For instance, the Location data element includes LocationCity, LocationState, LocationCountry, and LocationStatus. The SEARCH[Location] operator limits the search to the Location data element.

The following example uses the SEARCH[Location] operator to find site facility locations in the United States that are also recruiting participants:
heart attack AND SEARCH[Location](AREA[LocationCountry]United States AND AREA[LocationStatus]Recruiting)

In contrast, the following search expression, which does not use SEARCH[Location], would also return studies listing facility locations in the United States that are not recruiting and facility locations not in the United States that are recruiting:

heart attack AND AREA[LocationCountry]United States AND AREA[LocationStatus]Recruiting

Source operators

Source operators find studies similar to search terms. All source operators have the same precedence as a search term.

Operator	Example	Description
MISSING	AREA[ResultsFirstPostDate]MISSING	Finds study records that have no values in the search area specified as a parameter.
RANGE	AREA[ResultsFirstPostDate]RANGE[01/01/2015, MAX]	Finds study records in the search area that have a value greater than or equal to the first parameter (e.g., 01/01/2015) and less than or equal to the second parameter (e.g. MAX) in the search area. Each search area (e.g., ResultsFirstPostedDate) has a specified ordering, as follows: • Fields containing numbers and dates—Ordered in ascending sequence. • Age fields—Special ordering to accommodate units and to treat 1 year as greater than 6 months.

Operator	Example	Description
		Special RANGE operator values: • MIN—Use to indicate the smallest value of interest in the current search area. • MAX—Use to indicate the largest value of interest in the current search area. Note: The RANGE operator will not retrieve study records with no values in the search area.
ALL	ALL	Retrieves all study records in the database.

Scoring operators

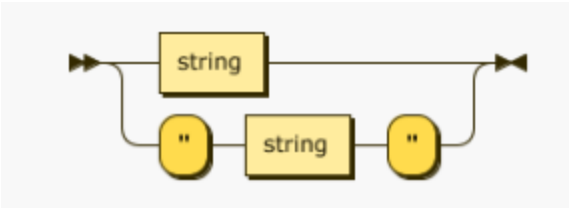
Scoring operators are used to adjust the rank order of search results by the most recent using dates or size of values.

Operator	Example	Description
TILT	TILT[StudyFirstPostDate]"heart attack"	Biases the scoring and rank ordering of study records in favor of the subexpression to the right by imposing a scoring penalty based on the ordering of API field values for the search area provided as a parameter (e.g., StudyFirstPostDate), with higher-ordered values having a lower penalty (e.g., more recent dates) than lower-ordered values (e.g., earlier dates). Use the TILT

Operator	Example	Description
		operator with API fields that are ordered, such as date fields.

Search terms

Search terms are words or phrases that must appear as values in the study records returned by the search. A search term consists of a string of characters without embedded search operators (https://clinicaltrials.gov.#search-operators). Enclosing a multi-word term in quotation marks indicates that all of the words in the search term must be found together in that order (that is, as a phrase) for a study record to be retrieved by the search (see below).



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To search for an operator as part of a term, add a backslash before the operator (e.g., \MISSING). If the operator used is quotation marks, then only embedded quotes need to be preceded by a backslash (e.g., \"sleep deprivation"\).

Search expressions

A search expression consists of sequences of search terms and operators that are evaluated by the search engine to find lists of studies. The search operators (https://clinicaltrials.gov.#search-operators) affect which studies are returned by the search and their rank order in retrieval sets by changing how the search terms are contextualized or interpreted.

All search expressions are OR expressions (https://clinicaltrials.gov.#orExpr).

Order of precedence among expressions

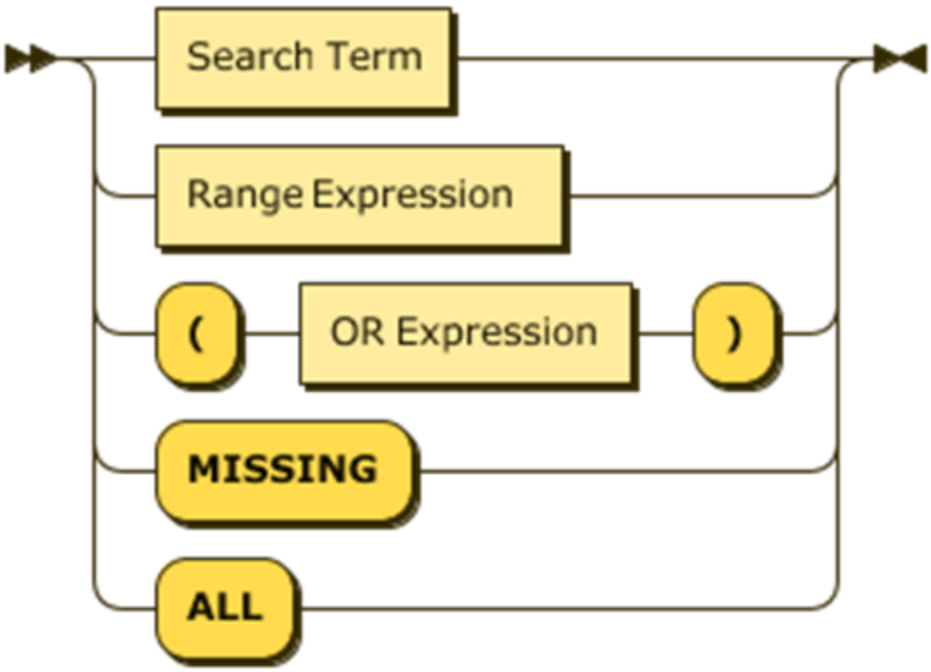
The order in which different types of search expressions are evaluated by the search engine is presented below from highest to lowest precedence. To

increase the precedence of an expression, enclose that expression in parentheses.

1. Source expression (https://clinicaltrials.gov.#sourceExpr).
2. Operator expression (https://clinicaltrials.gov.#operatorExpr).
3. AND expression (https://clinicaltrials.gov.#andExpr).
4. OR expression (https://clinicaltrials.gov.#orExpr).

Source expression

A source expression consists of a search term (https://clinicaltrials.gov.#search-terms), a range expression (https://clinicaltrials.gov.#rangeExpr), an OR expression (https://clinicaltrials.gov.#orExpr) enclosed in parentheses, and either the MISSING (https://clinicaltrials.gov.#missingOp) or ALL (https://clinicaltrials.gov.#allNoneOps) source operator (see below). The MISSING operator can be used for Structure type fields.



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The following examples are all source expressions:

- exhaustion
- RANGE[21 years,MAX] or RANGE[2025-03-20, 2025-03-31]
- (exhaustion OR sleep deprivation)
- MISSING

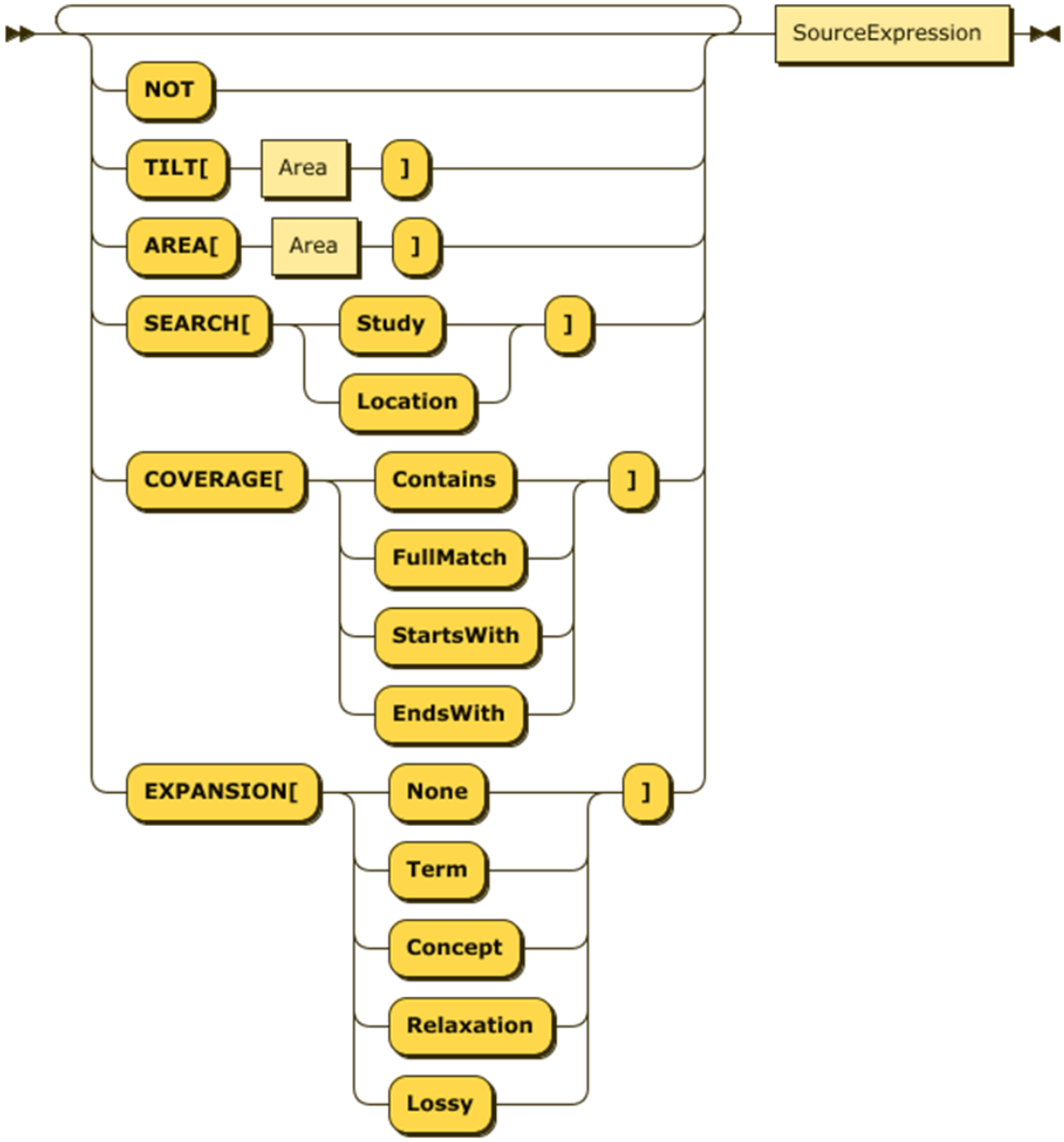
Range expression

A range expression, a type of source expression, consists of a RANGE (https://clinicaltrials.gov.#rangeOp) operator with minimum and maximum values as parameters. Use MIN to indicate the smallest value of interest in the search area and use MAX to indicate the largest value of interest in the search area. The parameters of a range expression may include dates, numbers, or numbers with units, such as with age. See examples above.

Studies with no values in the search area will not be retrieved by a search that uses a range expression.

Operator expression

An operator expression consists of a sequence of one, or more unary operators (e.g., the NOT (https://clinicaltrials.gov.#notOp) operator and all context operators (https://clinicaltrials.gov.#contextOpr)), followed by a source expression (https://clinicaltrials.gov.#sourceExpr). Any number of operator expressions may precede a source expression (see below).



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The TILT (https://clinicaltrials.gov.#tiltOp) and AREA (https://clinicaltrials.gov.#areaOp) operators take search areas (https://clinicaltrials.gov/data-api/about-api/search-areas) (labeled "Area" in the figure above) as a parameter. Note that the parameter is enclosed in square brackets, and there is no space between the operator and the left bracket.

The SEARCH (https://clinicaltrials.gov.#searchOp) operator takes either "Study" or "Location" as a parameter, as shown in the figure above.

The COVERAGE (https://clinicaltrials.gov.#coverageOp) and EXPANSION (https://clinicaltrials.gov.#expansionOp) operators take one of a small set of choices as a parameter (shown in the figure above). The COVERAGE operator takes Contains, FullMatch, StartsWith and EndsWith.

EXPANSION[Concept] and EXPANSION[Relaxation] can be used in combination with COVERAGE[Contains]. EXPANSION[Lossy] cannot be used with a COVERAGE operator.

The following examples are all operator expressions that include source expressions:

exhaustion

NOT **exhaustion**

EXPANSION[Concept] NOT (**exhaustion OR sleep deprivation**)

heart attack AND SEARCH[Location](AREA[LocationCity] Bethesda AND AREA[LocationState] Maryland)

These unary operators have the second highest precedence after source expressions.

AND expression

An AND expression consists of a list of one or more operator expressions (https://clinicaltrials.gov.#operatorExpr), separated by one or many binary Boolean AND (https://clinicaltrials.gov.#andOp) operators (see below).

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The following examples are all AND expressions that include operator expressions:

- **dizziness** AND **NOT exhaustion**
- **dizziness** AND **NOT exhaustio** AND **stroke**

The AND operator has the second lowest precedence among search operators. All operator expressions are evaluated before the AND expressions formed by connecting the operator expressions using AND operators.

OR expression

An OR expression consists of a list of one or more AND expressions (https://clinicaltrials.gov.#andExpr) (such as a search term (https://clinicaltrials.gov.#search-terms) or phrase) separated by binary Boolean OR operators (see below).

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The following examples are all OR expressions that include AND expressions:

- **heart attack** OR **stroke**
- **heart attack** OR **stroke** OR **dizziness AND shortness of breath**

The OR operator has the lowest precedence among search operators. All AND expressions are evaluated before the OR expressions formed by connecting the AND expressions using OR operators.

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